

MCS3301 – Control Engineering I

Dr. Ado Haruna

Dept. of Mechatronics Engineering

Bayero University, Kano

aharuna.mct@buk.edu.ng

Course Outline

- Concepts of feedback control
- Review of DEs and Laplace transforms
- Mathematical Models of Physical Systems
- Block Diagrams, Signal Flow Graphs, Analysis and Design in the S-plane, Steady state and transient response to step and ramp inputs.
- First Order Systems, Second order systems
- Use of PID lag, lead compensators
- Output Error Damping
- Stability Analysis

Assessment

- Group Assignment 10%
- Matlab Practical Sessions 10%
- Test 10%
- Exam 70%

References

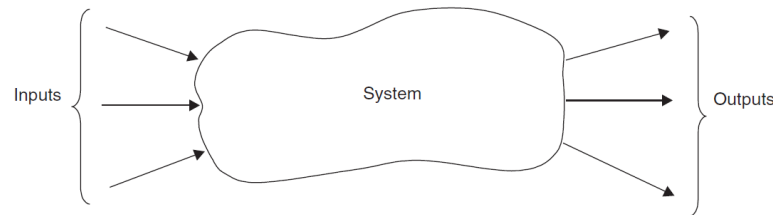
- Kuo, Benjamin C., and Farid Golnaraghi. *Automatic control systems*. 9th Edition. Englewood Cliffs, NJ: Prentice-Hall, 2009
- Dorf, Richard C., and Robert H. Bishop. *Modern control systems*. Pearson, 2011.
- Burns, Rowland S., *Advanced control engineering*, Butterworth-Heinemann, Oxford, 2001.

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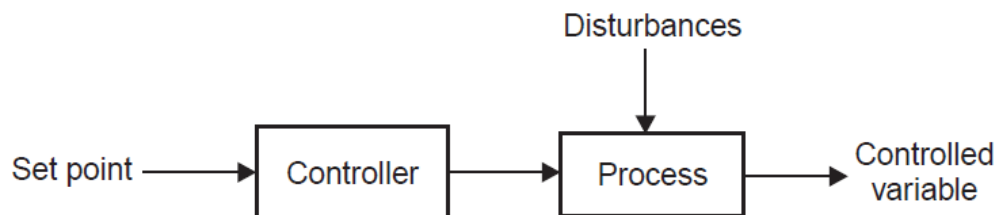
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Control Systems Components

- **Control system** engineers are concerned with **understanding** and **controlling** segments of their environment, often called **systems**.
- A **control** system is an interconnection of components forming a system configuration that will provide a **desired** output response.
- The **system** or **process** to be controlled is characterized by input(s) and output(s) as shown below.
- Control engineering, by its very nature is a **multidisciplinary** subject.

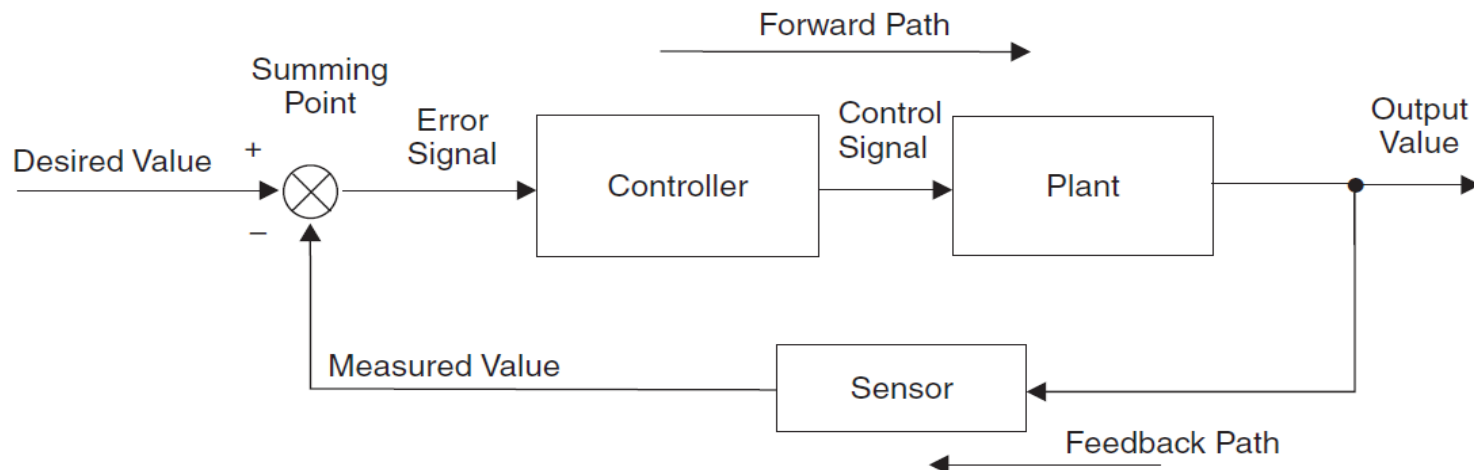


- A control system may either be an **open loop** or closed **loop system**.
- An **open loop** control system utilizes an **actuating** device to control the process directly **without** using feedback.



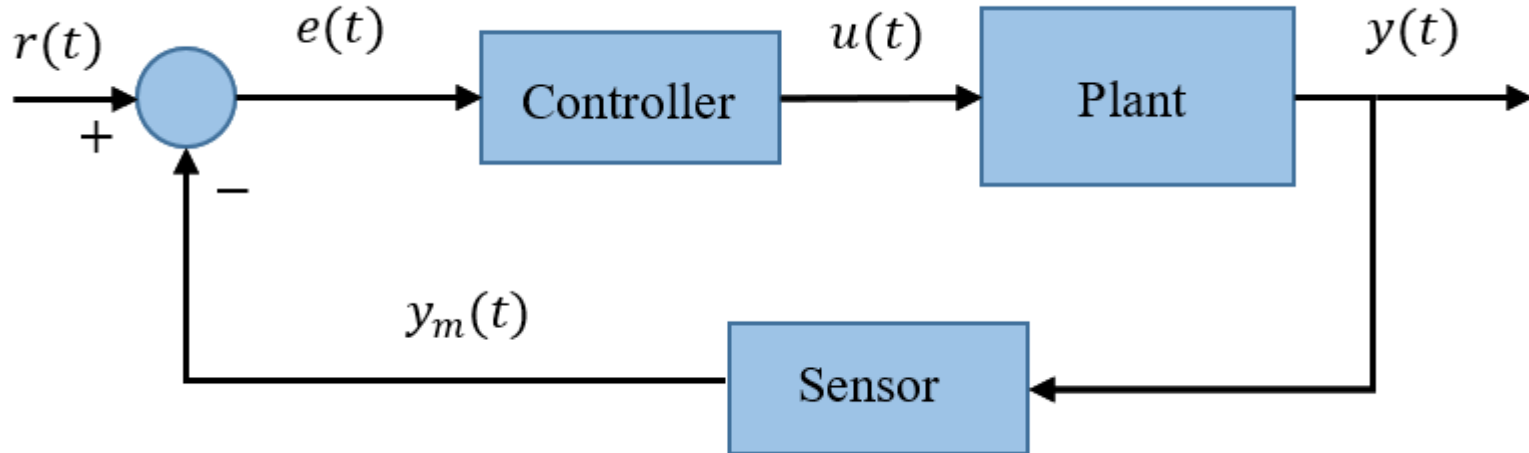
Control Systems Components

- A **closed-loop** control system utilizes an additional measure of the actual output known as a feedback signal to **compare** the actual output with the **desired** output response.
- A **feedback control** system thus maintains a prescribed relationship of one system variable to another by comparing functions of these variables and using the difference as a means of control.
- A closed-loop control has many advantages over open-loop control including the ability to **reject external disturbances**.
- The basic components in a feedback loop include the **plant**, **sensor**, **setpoint** and the **controller** as shown below.



The Closed-loop Framework

- The basic components in a feedback control structure are linked as shown below



where

$r(t)$ = the set point, reference or desired value;

$e(t)$ = the tracking error;

$u(t)$ = the control output and the input to the plant;

$y(t)$ = the system output;

$y_m(t)$ = the measured output.

- The tracking error is defined by $e(t) = r(t) - y_m(t)$

The Feedback Principle

- Control engineering is based on the foundations of feedback theory.
- Feedback controllers are designed to ensure that
 - 1) The closed loop system is **stable**;
 - 2) The output **tracks** the set point;
 - 3) The output is minimally affected by any possible **disturbances**
- The feedback principle can be illustrated in the following two cases;

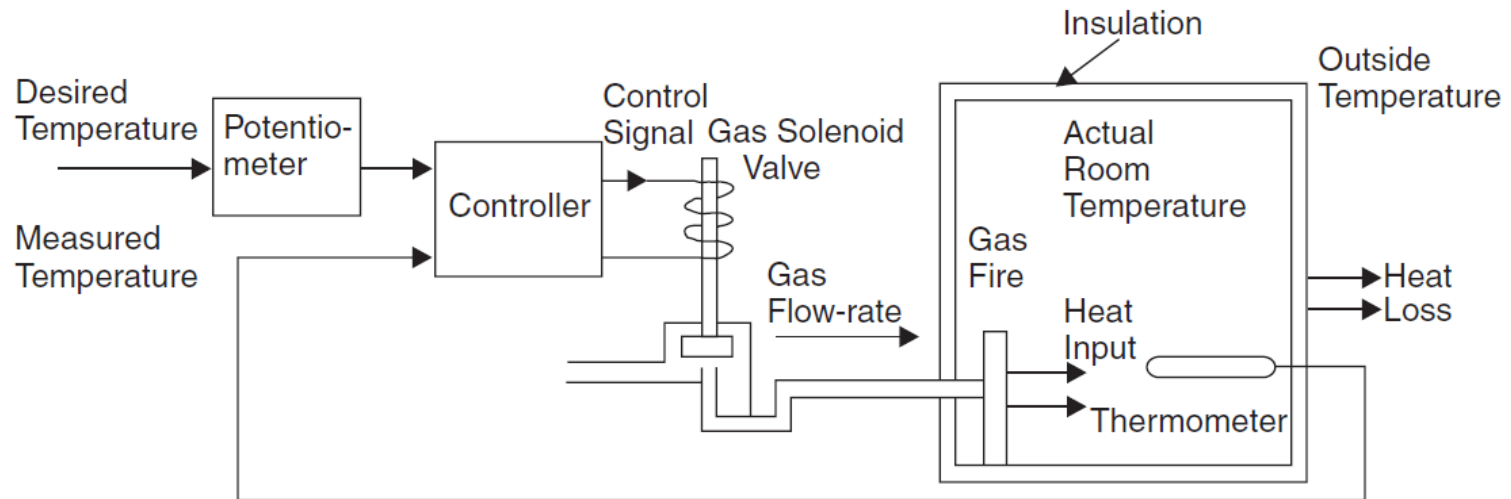
CASE1: If the setpoint is larger than the measured output of the system, then the tracking error will be **positive**, this leads to the increase of the system output and the measured output of the system. As a result the tracking **error** will **decrease**.

CASE2: If the setpoint is less than the measured output of the system, then the tracking error will be **negative**, this leads to the decrease of the system output and the measured output of the system. As a result the tracking **error** will **decrease**.

- Therefore, the **absolute** value of the tracking **error** will **decrease** if the set point is **not** equal to the measured output of the system.

Examples of Feedback Control Systems

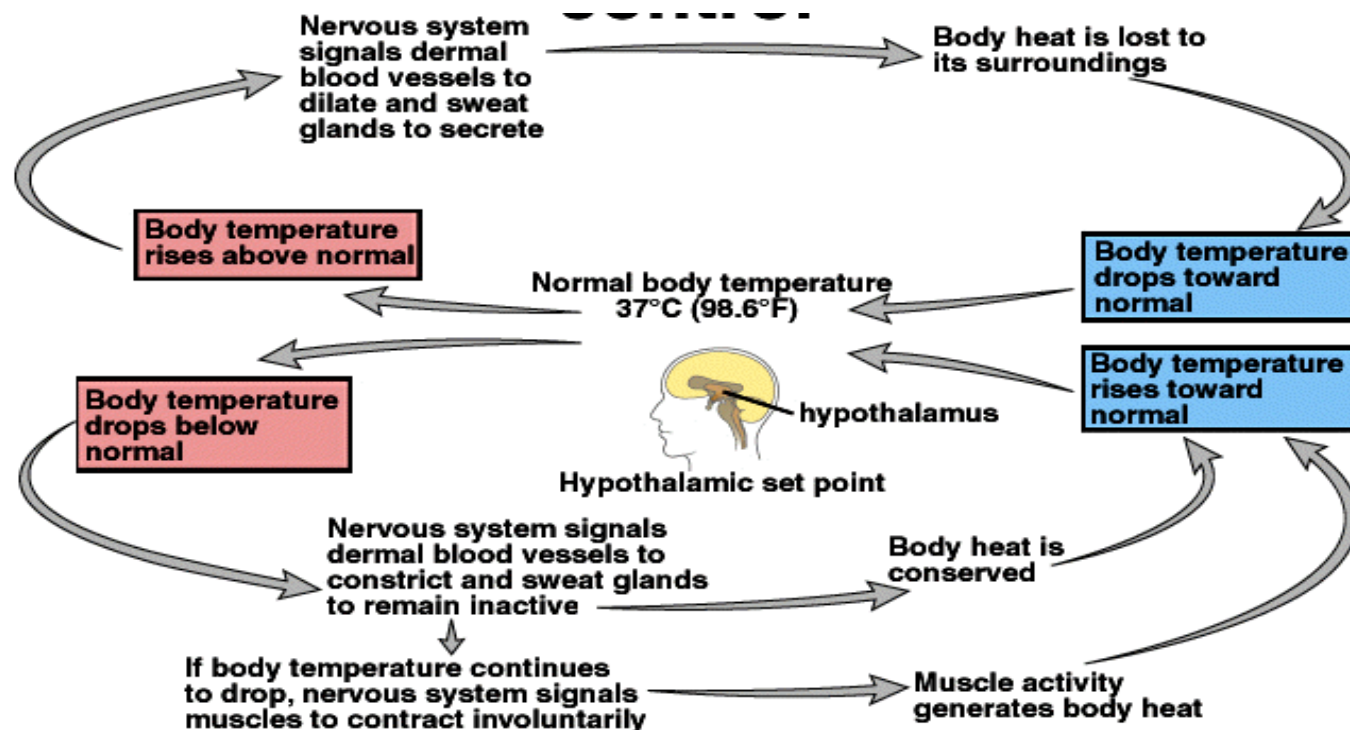
- The temperature is measured by a temperature **sensing** device.
- This is **compared** with the **desired** temperature setting on the potentiometer.
- Any difference or **error** causes the controller to send a **control signal** to the solenoid valve.
- This adjusts the **flow** of gas to the burner thereby **regulating** the temperature to the desired level.



Room temperature control system

Examples of Feedback Control Systems

- Regulation of human body temperature is an example of a feedback system.
- The human body temperature is kept constant at approximately 37 °C irrespective of the surrounding temperature.



Human body temperature regulation system